NYC 311

Service Demand & Operational Efficiency

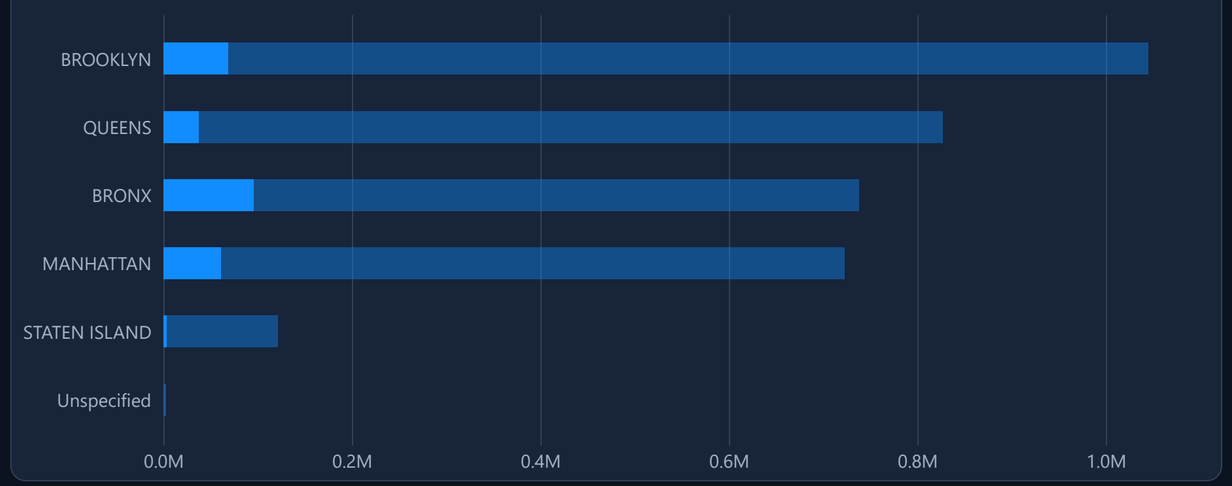
2024 | CITYWIDE SERVICE REQUEST ANALYTICS

Total Requests	Closure Rate	Average Resolution Days	Median Resolution Days	Non-Closed Status Requests	Missing Closure Date Requests
0M	98.88%	1.77	1.52	3K	3K

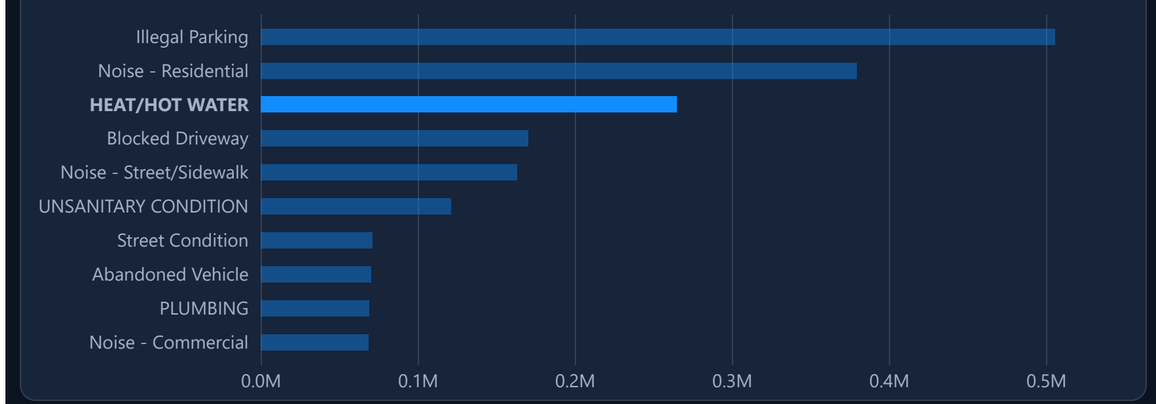
Monthly Service Request Volume



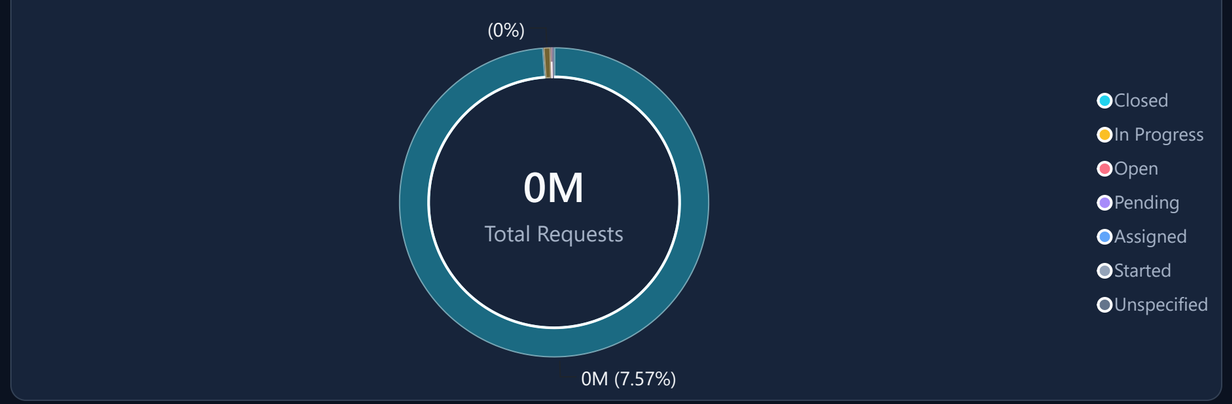
Service Requests by Borough



Top 10 Service Request Categories



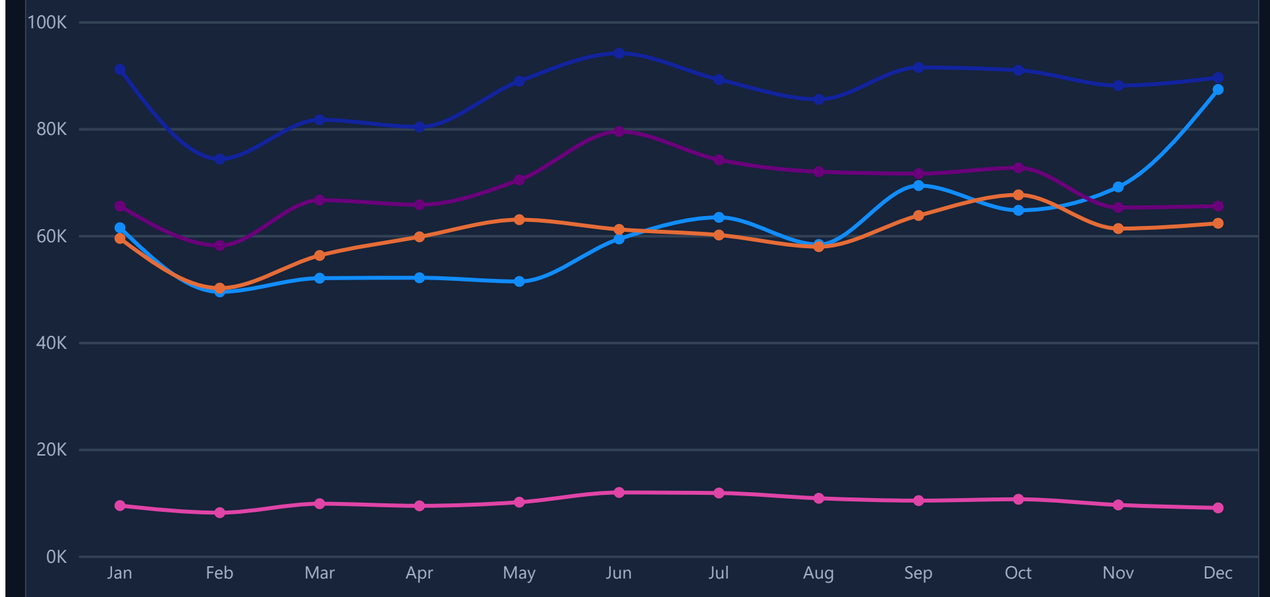
Request Status Distribution



SERVICE DEMAND ANALYSIS

Understanding request patterns across NYC boroughs, categories and seasons

Monthly Service Demand by Borough



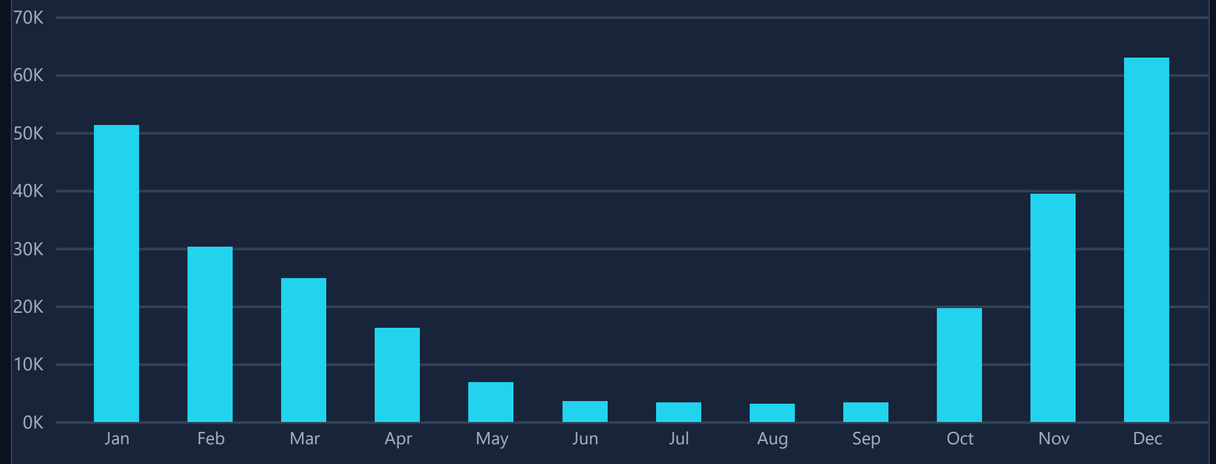
Select Borough

Select borough

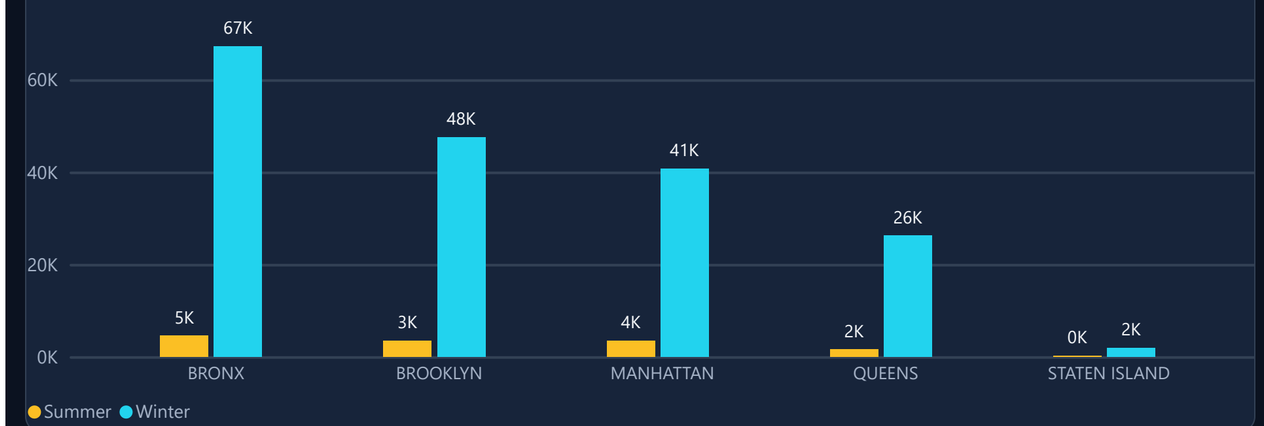
complaint_type

Select complaint_type

Monthly Heat & Hot Water Demand



Heat & Hot Water Demand: Winter vs Summer



Winter vs Summer Heat & Hot Water Demand by Borough

borough	Summer	Winter	Total
BRONX	4543	67311	**71854**
BROOKLYN	3464	47618	**51082**
MANHATTAN	3535	40798	**44333**
QUEENS	1593	26241	**27834**
STATEN ISLAND	213	1959	**2172**
Total	**13348**	**183927**	**197275**

KEY BUSINESS INSIGHTS

1. Seasonal Demand
Heat & Hot Water requests show a strong seasonal pattern, with demand increasing significantly during winter months.

2. Borough Differences
Winter demand is higher than summer demand across all five boroughs, although the scale of demand varies by borough.

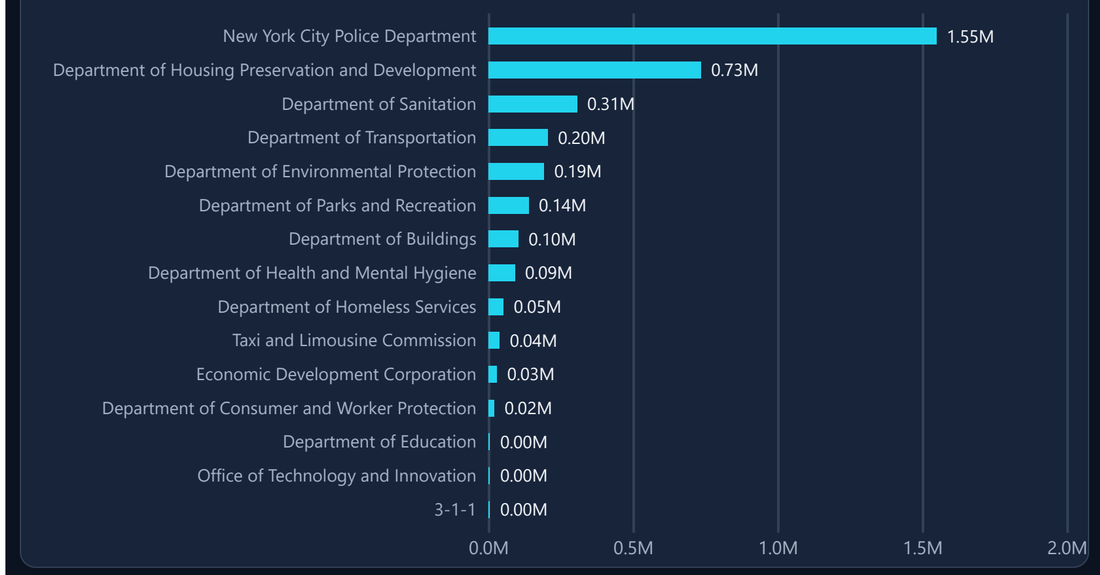
3. Complaint Concentration
Heat & Hot Water is the largest HPD complaint category across every borough.

4. Operational Planning
The seasonal demand pattern suggests that winter preparedness and resource planning should account for increased heating-related service requests.

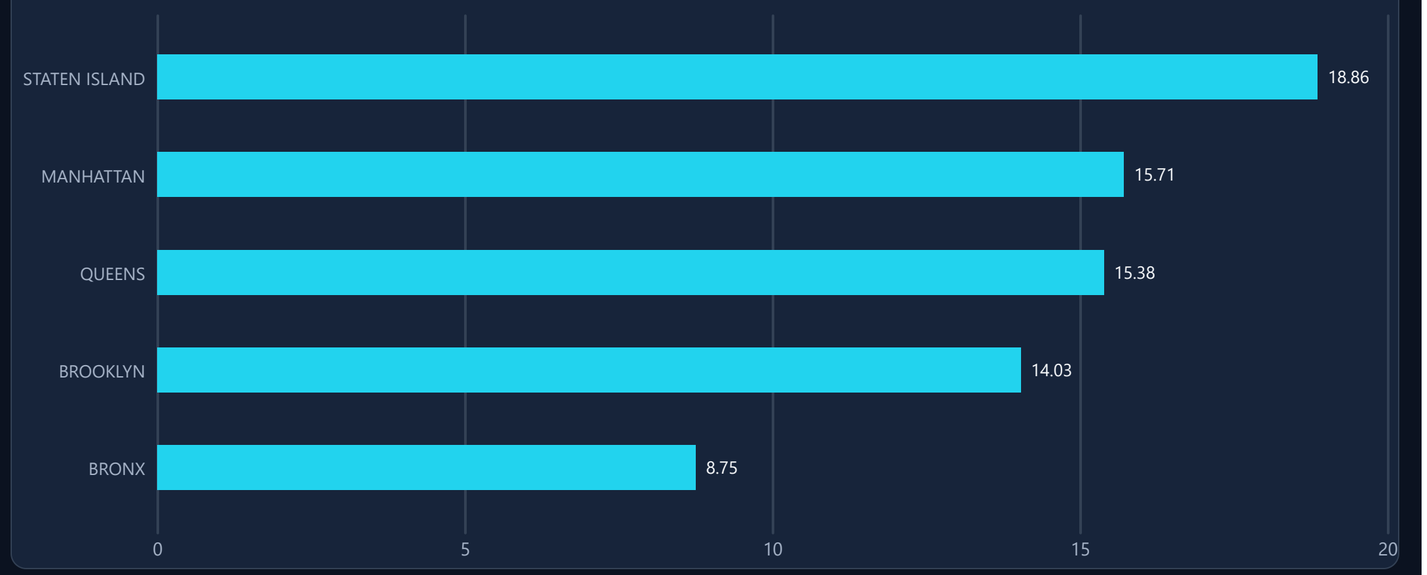
Operational Efficiency

Evaluating agency workload, service resolution patterns and operational performance

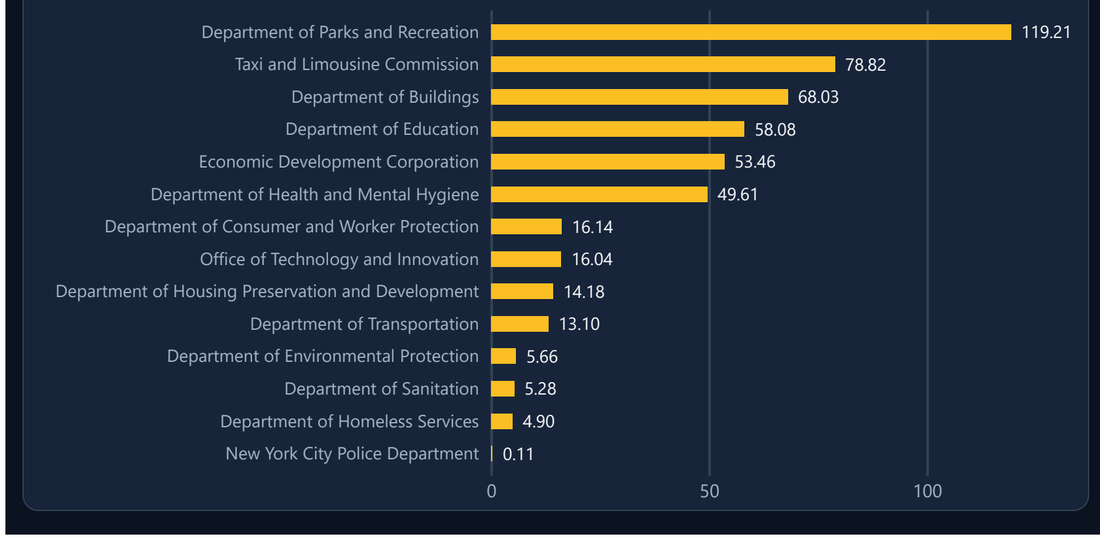
Agency Workload Distribution



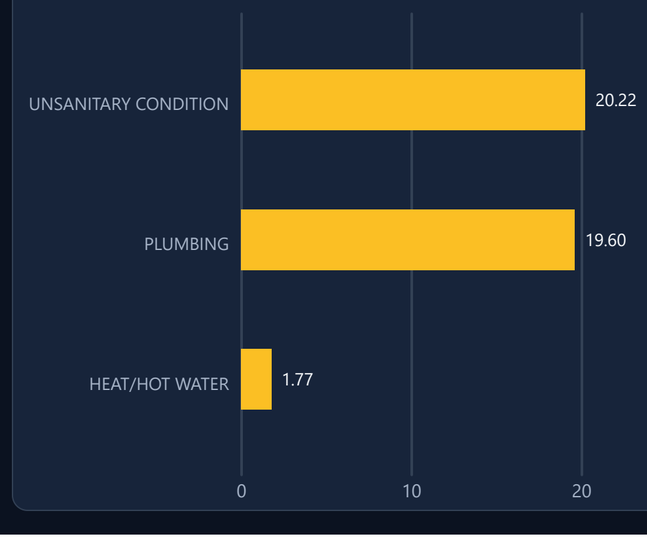
Average Resolution Time by Borough



Average Resolution Time by Agency



HPD Resolution Time by Complaint Type



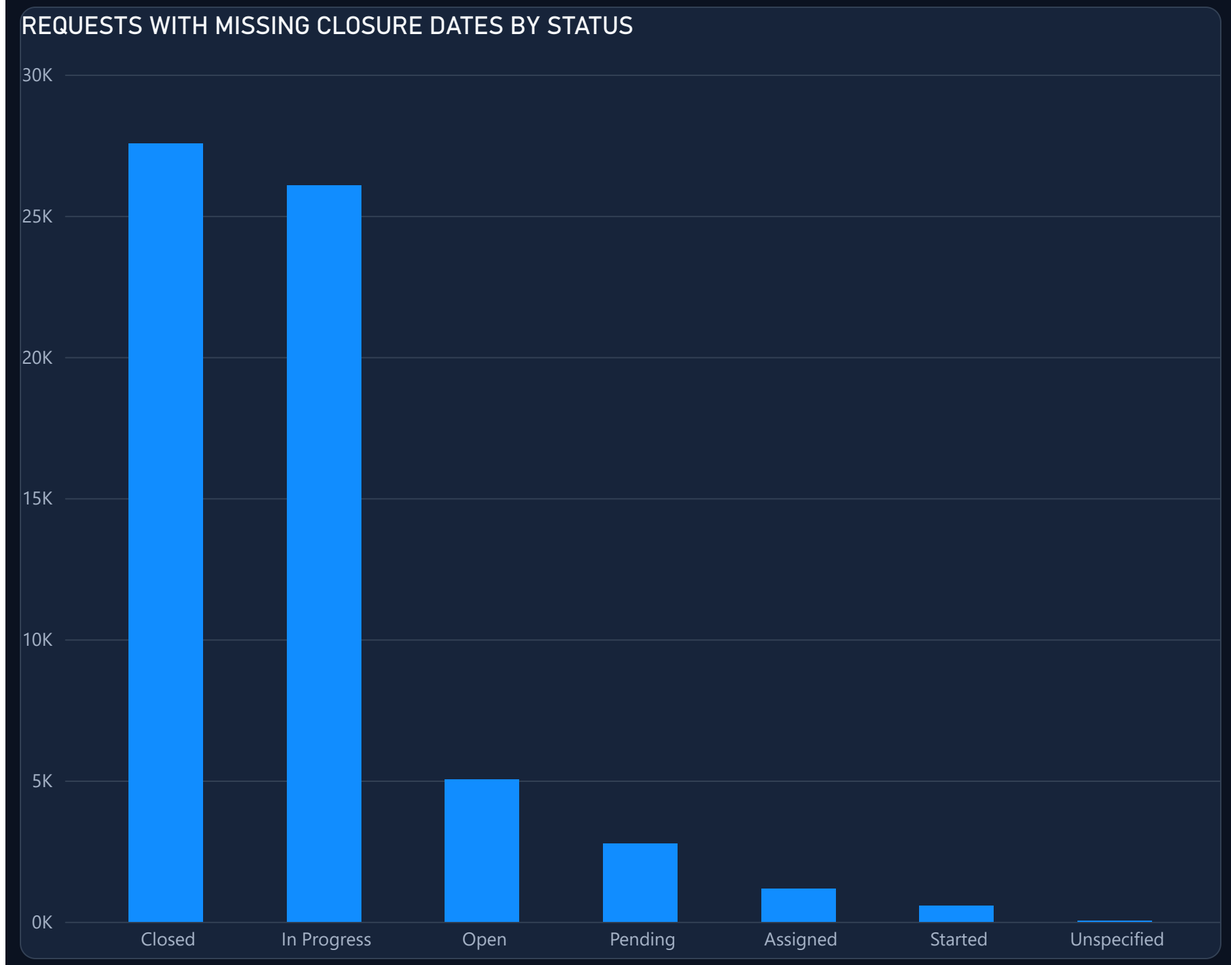
HPD Resolution Time by Complaint Type and Borough

complaint_type	BRONX	BROOKLYN	MANHATTAN	QUEENS	STATEN
UNSANITARY CONDITION	16.64	20.54	27.48	18.48	
PLUMBING	17.83	19.68	24.14	17.49	
HEAT/HOT WATER	1.49	1.92	2.09	1.69	
Total	**7.75**	**10.25**	**11.27**	**8.50**	

DATA QUALITY & BUSINESS INSIGHTS

Validating source data, identifying inconsistencies and translating findings into actionable recommendations

Closed Status Missing Date	Non-Closed With Closure Date	Valid Resolution Records	Missing Closure Date Requests	Missing Resolution Records
28K	3K	3M	63K	64K



KEY DATA QUALITY FINDINGS

1.Closure Date Gaps: Some requests marked as Closed have no recorded closure date, affecting resolution-time reporting.
2.Status Inconsistencies: Some requests with non-closed statuses contain closure dates, indicating records that may need validation.
3.Resolution Data Coverage: Missing resolution-time values limit the ability to assess service resolution consistently.
4.Reporting Recommendation: Standardize status and closure-date validation to improve operational reporting and data reliability.

